COM-480: Milestone 2

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1 Sketches

The first part of the website is sequential: the user must press the screen to go to another page (e.g. clicking the 'World Values Survey' logo will bring back to the menu, pressing an answer to a question brings the next question, etc.).

However, the *Explore* section is one long scrollable page.

1.1 Website menu

The menu is shown in Fig. 1a, the user can either choose a global survey encompassing questions from all categories, or choose a specific category they wish to focus on. The globe on the left of the page is meant to rotate.

1.2 Questions page

Once the user has selected the survey they wish to respond to, the questions appear as shown in Fig. 1b.

When the user brings their mouse to a possible answer, the bar and number are highlighted to show the answer they are selecting.

As for the menu, the globe on the left should rotate while the questions are being answered. The globe is simultaneously a choroplet map, where darker countries matching the user better. The map colors changes as questions are answered, making the experience more fun and interactive.

1.3 Visualization page

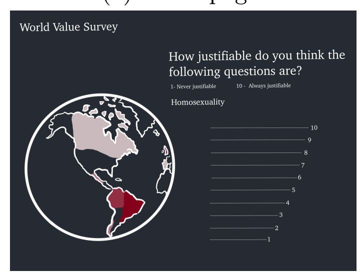
Once all the questions have been answered, the globe slowly grows to fit the page, and the country that fits the user the most is displayed on top of the globe, with the percentage of similarity displayed.

If the user presses on a country on the rotating globe, they can have more detailed information on the country's inhabitants' answers, focusing e.g. more on age, gender, religion, ethnicity or economical class, as depicted in Fig. 1c. This feature aims at giving users a chance to focus e.g. on the answers of a specific group of people within the country, and not just follow the mean (a user might e.g. care more for the young population's opinion rather than all age groups).
For each selected country, the '% match' is also displayed, even if the country is not the best fit according to our algorithm.

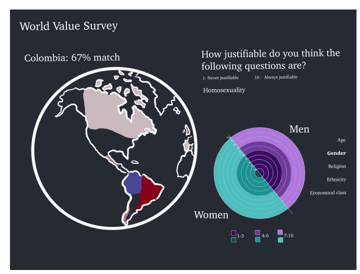
To visualize the answers of specific groups, we first use stacked bar charts, but slightly change the classical straight display into a round representation (Fig. 1c). If this is too hard to implement, we will take a simple classical stacked bar chart representation.

For other questions, we use wordclouds as a second representation method (Fig. 1d).

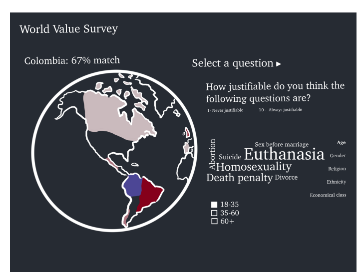
(a) Menu page.



(b) Questions page.



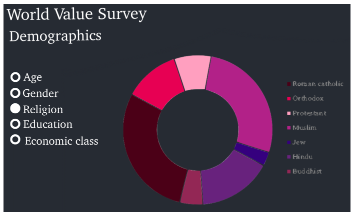
(c) Visualization page using round stacked bar chart.



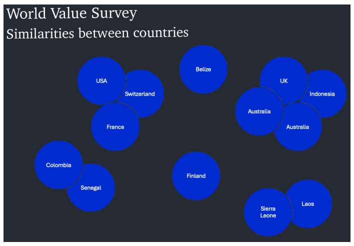
(d) Visualization page using wordclouds.

Figure 1: Website sketch: Surveys.

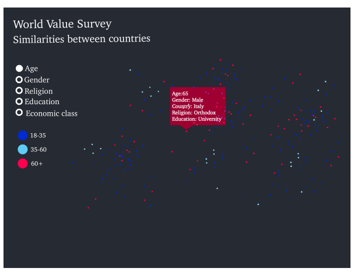
1.4 Explore page



(a) Demographics visualization.



(b) Visualize similarities between countries.



(c) Visualize individuals as clusters.

Figure 2: Website sketch: Explore page.

We added an *Explore* page to gain insight into the dataset. The insights are not personal to the user; however, some visualizations are interactive and therefore allow the user to choose specifically what information they wish to have.

In Fig. 2a, we show the different groups that responded to the survey.

As for Fig. 2c, we show the clusters formed by countries when taking the overall mean of answers.

Finally we also plot as small dots all individuals that responded to the survey, and we allow the user to select either 'Age', 'Gender', 'Religion', 'Economic Class', 'Education', 'Continent', to highlight the dots as different categories. This is meant as a conclusion of the experience, to highlight the diversity in the values; we expect clusters of individuals to not necessarily be from the same continent/region.

2 Tools

The website's visualizations are built with the following JavaScript libraries, some drawn from specific course lectures, others fetched by ourselves:

- Rotating and choropleth globe:
  - **amCharts** library
  - **D3.js** (Lecture 4.2) [1]
  - **d3-tip** (Lecture 5.1) [2]
- Bar chart (Fig. 1c):
  - **D3.js**: Radial stacked bar chart (Lecture 11) [3] [4]
- Word Cloud (Fig. 1d):
  - **d3-cloud** / **amcharts**: Tag cloud. (Lecture 9) [5] [6]
- Pie chart (Fig 2a):
  - **Charts.js** [7]
- Clusters (we will use t-SNE and then use scatterplots) (Figures 2b and 2c):
  - **D3.js** / **amCharts**: Scatterplot (Lecture 4.2) [1] [8] [9]

3 Extra ideas

We would like to add in the *Explore* section a graph showing the answers to two questions simulatenously (one question displayed on the x-Axis, the other on the y-Axis). The idea is to give the user the possibility to choose the questions on each axis, and then visualize groups of people (e.g. by 'Religion', 'Age', 'Economic class', etc.). The groups would then be represented as bubbles, highlighting how different groups' opinions vary.

The second idea we have is to do a Sankey plot. The user selects the gender, ethnicity, continent, age, economic class, and education level, and the final output is the probability of fitting their own country. That is, we would have to, for each individual who responded to the survey, measure how much they match their own country, and then use the Sankey plot to visualize the data. This might be useful in anticipating migratory fluxes; some groups of people might be typically unaligned with the overall values of their country / continent.

References

[1] K. Benzi, “Data-driven documents,” 2024. Lecture 4.2, EPFL.

[2] K. Benzi, “Interactions & views,” 2024. Lecture 5.1, EPFL.

[3] K. Benzi, “Tabular data,” 2024. Lecture 11, EPFL.

[4] Y. Holtz, “Circular barplot.” https://d3-graph-gallery.com/circular_barplot.html, 2025. Accessed: April 17, 2025.

[5] K. Benzi, “Text visualization.” Lecture 9, EPFL, 2024. Based on “Text Visualization” slides.

[6] amCharts, “Word cloud.” https://www.amcharts.com/demos/word-cloud/, 2025. Accessed: April 17, 2025.

[7] Chart.js Contributors, “Chart.js.” https://www.chartjs.org/, 2025. Accessed: April 17, 2025.

[8] d3-shape contributors, “d3-shape: Arcs and pie charts.” https://github.com/d3/d3-shape, 2025. Accessed: April 17, 2025.

[9] amCharts, “Donut chart.” https://www.amcharts.com/demos/donut-chart/, 2025. Accessed: April 17, 2025.